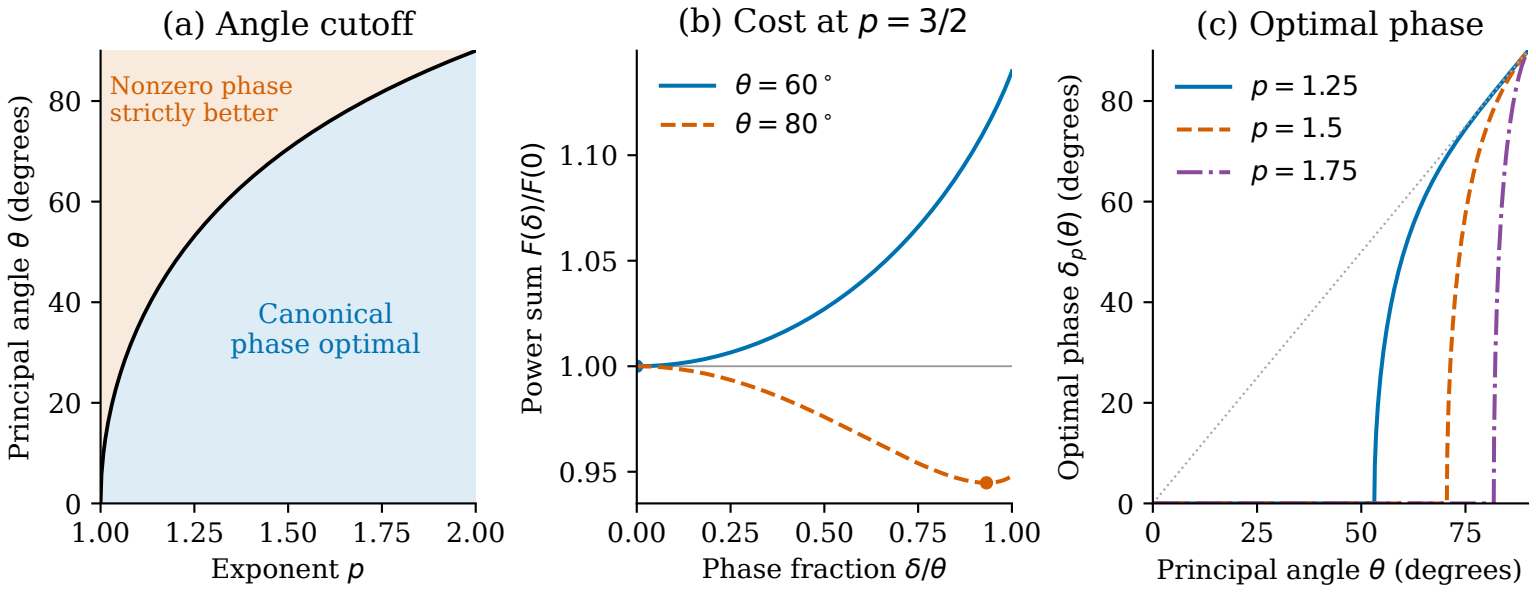
(a) Angle cutoff
Nonzero phase
strictly better
Canonical
phase optimal
Principal angle θ (degrees)
Exponent p
1.00
1.25
1.50
1.75
2.00
0
20
40
60
80
(b) Cost at p = 3/2
θ = 60°
θ = 80°
Power sum F(δ)/F(0)
Phase fraction δ/θ
0.00
0.25
0.50
0.75
1.00
0.95
1.00
1.05
1.10
(c) Optimal phase
p = 1.25
p = 1.5
p = 1.75
Optimal phase δ_p(θ) (degrees)
Principal angle θ (degrees)
0
25
50
75
0
20
40
60
80